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Effects of wet torrefaction on the physicochemical properties and pyrolysis product properties of rice husk

Shuping Zhang [a], Tao Chen [a], Yuanquan Xiong [a,*], Qing Dong [b]

[a] Key Laboratory of Energy Thermal Conversion and Control of Ministry of Education, School of Energy and Environment, Southeast University, Nanjing 210096, China
[b] Faculty of Life Science & Food Engineering, Huaiyin Institute of Technology, Huaian, China

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ABSTRACT

Wet torrefaction in the temperature range from 150 to 240 °C for 60 min was applied on rice husk sample prior to pyrolysis process in this work. Its effects on the physicochemical properties and pyrolysis product properties were evaluated. The analysis results of physicochemical properties of raw and pretreated samples indicated that wet torrefaction not only improved the fuel characteristics but also removed a large amount of alkali and alkaline earth metal species with the dual advantages of dry torrefaction and demineralisation. On the basis of pyrolysis experiment results, it was found that wet torrefaction at the mild reaction condition had a positive impact on bio-oil production, but more severe wet torrefaction condition was not reasonable. The highest relative content of levoglucosan (57.2%) was obtained from pyrolysis of rice husk sample after wet torrefaction at 210 °C, which was about 6.2 times higher than that of raw rice husk. The results of biochar analysis suggested that biochar produced from sample after wet torrefaction and pyrolysis seems not a desirable solid fuel, it can be used as the feedstock for preparation of nanosilica.

1. Introduction

Pyrolysis has been considered as a processing technology to convert biomass into desirable bioenergy in an eco-friendly and cost-effective manner, such as non-condensable gas, bio-oil and biochar [1–3]. Bio-oil, a complex liquid mixture, can be directly applied for heat and power production, or used for preparation of transportation fuels with upgrading processes [4,5]. However, high water content, high oxygenated organic compounds, low heating value, acidity and chemical instability of bio-oil obtained from pyrolysis of raw biomass hampered the efficient use of bio-oil. In addition to bio-oil, biochar is also a value-added pyrolysis product, which can be applied as solid fuels, soil amendments or a feedstock of activated carbon [6–8]. However, raw biomass material with low energy density, low bulk density, high water content and poor fuel characteristics limited the development of biomass pyrolysis. The high inorganic minerals content contained in raw biomass also significantly lowered the quality of pyrolysis products [9,10]. Therefore, biomass pretreatment prior to pyrolysis is an effective approach to improve the quality of target products (bio-oil and biochar) obtained from biomass pyrolysis [11,12].

Recently, wet torrefaction, also called hydrothermal pretreatment or hot compressed water pretreatment, has attracted significant attention due to it is an effective thermochemical pretreatment process for biomass [13–15]. Wet torrefaction is carried out in hot compressed water in the temperature range of 150–260 °C and the process pressure is usually slightly higher than the saturated vapor pressure at the corresponding temperature [13]. Much information has been obtained about the wet torrefaction of biomass. Previous works mainly focused on the physicochemical properties of biomass feedstocks after wet torrefaction. For example, the heating value, energy density, hydrophobicity and grindability of biomass feedstocks can be improved by the process of wet torrefaction [16,17]. The effects of wet torrefaction on pyrolysis behavior and pyrolysis product properties of biomass have not been sufficiently evaluated. Bach et al. [18] studied the pyrolysis behavior and kinetic parameters of Norway spruce and birch woods under nitrogen atmosphere with thermogravimetric analyzer. Yan et al. [19] confirmed that wet torrefaction improved the quality of loblolly pine wood for subsequent thermochemical process using thermogravimetric analyzer. Chang at al. [14] found that wet torrefaction increased the yield and fuel quality of bio-oil obtained from pyrolysis of eucalyptus wood because of the selectively removal of hemicellulose and inorganic minerals by wet torrefaction. Le Roux et al. [20] demonstrated that wet torrefaction

* Corresponding author.
E-mail address: yqxiong@seu.edu.cn (Y. Xiong).

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had little influence on the bio-oil yield but improved the quality of bio-oil obtained from fast pyrolysis of aspen wood.

Rice husk is known to be one of the most widely agricultural wastes, which can be utilized as energy source. Approximately 150 MT of rice husk was generated globally in 2015 according to recent estimations [21]. Rice husk used as biomass pyrolysis feedstock to produce bio-oil and biochar has been reported by many reports [22–24]. Meanwhile, wet torrefaction of rice husk has also been evaluated in the previous work, the results indicated that the energy density was increased and the atomic O/C ratio was decreased after wet torrefaction [25]. Nevertheless, no published technical reports about the effect of wet torrefaction on the pyrolysis product properties of rice husk have been found.

In this work, the main objective is try to evaluate the feasibility of producing high quality bio-oil and biochar from by subjecting rice husk samples with wet torrefaction to modify the physicochemical properties of rice husk prior to pyrolysis. To this end, wet torrefaction in the temperature range from 150 to 240 °C for 60 min was applied on rice husk prior to pyrolysis process. Two parts of contents were analyzed to investigate its effects on the physicochemical properties and pyrolysis product properties of rice husk. The physicochemical properties of raw and pretreated samples were evaluated by proximate and elemental analyses, functional groups analysis and surface morphology analysis. In addition, the product yields and qualities of bio-oil and biochar produced from pyrolysis of raw and pretreated samples were analyzed to provide the fundamental data for biomass pyrolysis process.

2. Materials and methods

2.1. Raw materials

Rice husk was acquired from a rice milling factory in Jiangxi province of China. The biomass feedstock was screened to remove coarse contaminants prior to experiment. The dried rice husk (RH) was stored in sample bag after drying at 105 °C to constant weight.

2.2. Wet torrefaction

Wet torrefaction was conducted in a high-pressure batch reactor at 150, 180, 210, and 240 °C for 60 min as shown in Fig. 1. In a typical run, around 10 g of rice husk was mixed with 200 mL of deionized water was added into the 300 mL stainless steal autoclave, and then the reactor was set into a heating furnace with a 1.5 kW heating power. Prior to pretreatment experiment, nitrogen gas was used to eliminate the oxygen in the reactor. Reactor temperature and pressure was indicated by the k-type thermocouple and pressure gauge, respectively. When wet torrefaction was completed, the reactor was allowed to cool in an ice-water bath. The solid and liquid samples were separated by vacuum filtration. The pH value of liquid sample was measured. The obtained solid samples were washed several times with deionized water to bring the pH to neutral, and then oven dried overnight to achieve constant weight. Each wet torrefaction process was repeated three times under the same conditions to ensure experimental reproducibility. Wet torrefied rice husk samples were labeled as RHx, where x represents the different wet torrefaction temperature of 150, 180, 210 and 240 °C, respectively.

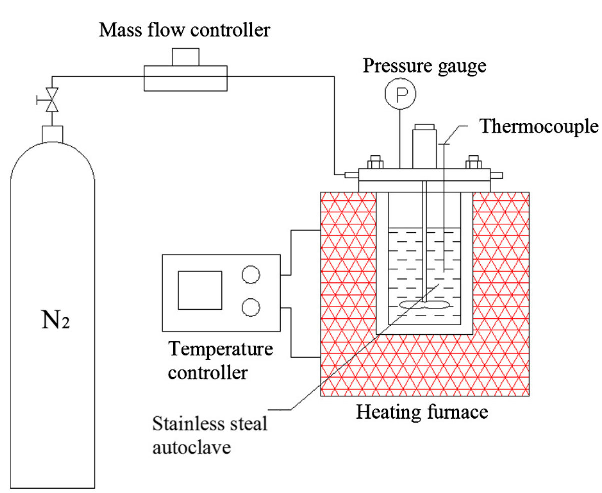

Fig. 1. A high-pressure batch reactor for wet torrefaction.

2.3. Characterization of raw and pretreated rice husk samples

The proximate analysis of raw and pretreated rice husk was determined in accordance with ASTM standards. The elemental analysis (C, H, and N) was conducted by the Vario EL-III elemental analyzer. The higher heating value (HHV) was measured by the SDACM3000 calorimeter. The mass yield, energy density and energy yield were calculated as follows:

$$\text{Mass yield} = \frac{m_{wt}}{m_r} \times 100\% \tag{1}$$

$$\text{Energy density} = \frac{HHV_{wt}}{HHV_r} \tag{2}$$

$$\text{Energy yield} = \frac{m_{wt}HHV_{wt}}{m_rHHV_r} \times 100\% \tag{3}$$

where m_r and m_{wt} mean the mass of raw and wet torrefied samples. HHV_r and HHV_{wt} mean the higher heating values of raw and wet torrefied samples, respectively.

An inductively coupled plasma optical emission spectrometry (ICP-OES) system (Leeman Labs Inc., USA) was used to analyze the concentration of alkali and alkaline earth metal species (AAEMs) in rice husk sample by microwave digestion with the HNO_3:HCl:H_2O_2 (8:1:1 v/v) mixture according to the method described in our previous work [24]. The FTIR spectra obtained from a Bruker Vector 22 Fourier transform infrared spectrophotometer in wave number range 400–4000 cm^{-1} with a resolution of 4 cm^{-1} was used to investigate the changes in functional groups by wet torrefaction. The surface morphologies of raw and pretreated samples were examined by a LEO 1530 VP scanning electron microscope (SEM).

2.4. Pyrolysis experiment

Pyrolysis of rice husk sample was carried out in the vertical drop fixed-bed reactor as shown in Fig. 2, which was described in our previous work [8,23]. Briefly, N_2 with 200 mL/min flow rate was used to provide an inert atmosphere. When the pyrolysis temperature of 550 °C was reached, rice husk sample of 5 g was inserted rapidly from the sample feeder into the quartz reactor. After holding for 10 min under the pyrolysis conditions, the quartz reactor was moved out from the heating furnace for cooling. After then, the biochar, bio-oil and non-condensable gas were collected from the quartz reactor, condenser and gas sampling bags, respectively. For mass balance calculation, the biochar and bio-oil yields were weighted and that of non-condensable gas was determined by difference. Three parallel experiments were performed for each reaction condition. The biochar obtained from pyrolysis of raw and pretreated samples are labeled as RHC, RHC150, RHC180, RHC210 and RHC240, respectively.

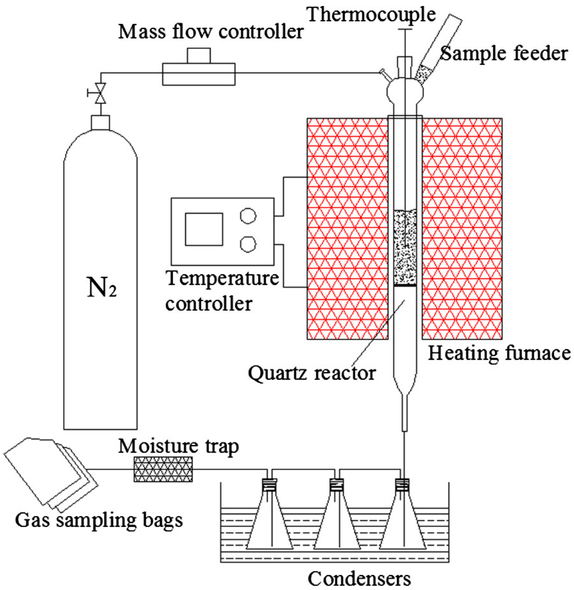

Fig. 2. A vertical drop fixed-bed reactor for pyrolysis.

2.5. Characterization of pyrolysis products

The main chemical species of bio-oil were determined using gas chromatography–mass spectrometry (GC–MS, Agilent 7890A/5975C). All the chromatographic peaks were identified by the database of the NIST-MS library. Detailed operating parameters for GC–MS operations can be obtained in our previous work [23,26]. The water content of bio-oil was measured by Karl-Fischer titration. The pH value of bio-oil was measured by a PHS-3C pH meter. The higher heating value of bio-oil was measured by the SDACM3000 calorimeter.

The fuel characteristics of biochar were also analyzed using the methods mentioned in Section 2.3. The chemical components of ash in biochar were determined by the ARL-9800 X-ray Fluorescence (XRF) spectroscopy.

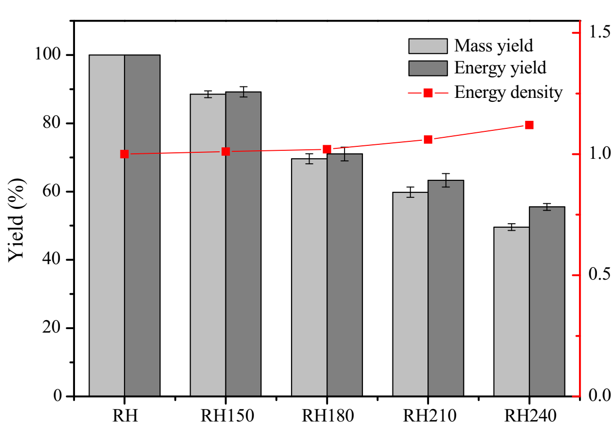

Fig. 3. Mass yield, energy yield (average of three replications ± standard deviations) and energy density of wet torrefaction.

3. Results and discussion

3.1. Effects of wet torrefaction on physicochemical properties

3.1.1. Mass and energy yields of wet torrefaction

The mass yield, energy yield and energy density of wet torrefaction are presented in Fig. 3. It can be seen that the mass yield decreased gradually from 86.7% to 47.9% as the wet torrefaction temperature increased from 150 to 240 °C. Similar tread can be observed for the energy yield during wet torrefaction. However, the value of energy yield is always higher than that of mass yield at the same pretreatment condition, which indicated that the energy density was enhanced by wet torrefaction. As shown in Fig. 3, the energy density increased with wet torrefaction temperature from a low of 1.01 for RD150 to a high of 1.12 for RH240. Generally, increasing severity of wet torrefaction resulted in reduction in mass and energy yields and enhancement of energy density, which was similar to the result reported by others [27].

3.1.2. Fuel characteristics

The results of fuel characteristics, such as proximate and elemental analyses and HHVs of raw and wet torrefied samples are shown in Table 1. Demineralization process can be achieved during wet torrefaction due to the decrease in ash content from 11.8% for RH to 10.9% for RH150. When the reaction temperature was increased from 150 to 240 °C, the ash content increased gradually from 10.9% to 15.5%. The results obtained from this work were consistent with the finding of other researchers, which indicated that the increase in the ash content with the increase in wet torrefaction temperature was the results of decomposition of organic components and the low solubility of rice husk ash during wet torrefaction [28,29]. At the mild wet torrefaction condition, the volatiles content increased significantly from 73.5% for RH to 76.9% for RH180. When the reaction temperature further increased to 240 °C, the volatiles content decreased to 63.0% for RH240. A converse trend for the effects of wet torrefaction condition on the fixed carbon content can be obtained. In addition, the elemental analysis results revealed that the carbon content slightly increased while the oxygen content gradually decreased with the increase in reaction temperature. It is noteworthy that nitrogen content decreased significantly from 1.2 for RH to 0.4 for RH240, which indicated that wet torrefaction is an efficient method for nitrogen removal. Meanwhile, the decreased O/C and H/C atomic ratios during wet torrefaction can also be observed from 0.74 and 1.68 for RH to 0.54 and 1.36 for RH240, respectively, which can be explained by the effect of dehydration, decarboxylation and demethanation reactions. In addition, the increase in HHVs with the increase in wet torrefaction temperature can also be obtained due to the increased carbon content in pretreated rice husk.

The concentration of typical alkali and alkaline earth metals (AAEMs) in raw and wet torrefied samples with different reaction temperature is compared as shown in Table 2. It can be observed that 60–90% of AAEMs can be removed by wet torrefaction. In addition, additional AAEMs can be removed from rice husk samples when the reaction temperature was increased. The pH value of the liquid sample produced from wet torrefaction was 5.0, 4.5, 4.3 and 3.7 in accordance with the reaction temperature of 150, 180, 210 and 240 °C, respectively. It was reported that some organic acids were enriched in the liquid sample produced from wet torrefaction, and it has the potential for removal of AAEMs [30]. Due to most ion-exchangeable AAEMs were bond to the organic acid functional groups existed in the extractives or hemicellulose, the removal efficiency of AAEMs was dependent on the degree of extractives and hemicellulose decomposition [31].

Table 1
Fuel characteristics of rice husk samples.

Samples	Proximate analysis (wt.%, db)			Ultimate analysis (wt.%, db)				Atomic ratio		HHV (MJ/kg)
	A_d	V_d	FC_d	C	H	O	N	O/C	H/C	
RH	11.8 ± 0.2	73.5 ± 0.5	14.7 ± 0.2	40.8 ± 0.4	5.7 ± 0.1	40.5 ± 0.3	1.2 ± 0.1	0.74	1.68	16.2 ± 0.1
RH150	10.9 ± 0.2	76.5 ± 0.4	12.6 ± 0.2	41.6 ± 0.2	5.7 ± 0.1	41.0 ± 0.4	0.8 ± 0.1	0.74	1.64	16.3 ± 0.1
RH180	12.1 ± 0.1	76.9 ± 0.4	11.0 ± 0.3	43.3 ± 0.2	5.7 ± 0.1	38.3 ± 0.3	0.6 ± 0.1	0.66	1.58	16.5 ± 0.1
RH210	12.8 ± 0.1	74.8 ± 0.3	12.4 ± 0.3	43.9 ± 0.1	5.5 ± 0.1	37.3 ± 0.4	0.5 ± 0.1	0.64	1.50	17.1 ± 0.1
RH240	15.5 ± 0.1	63.0 ± 0.4	21.5 ± 0.2	45.8 ± 0.2	5.2 ± 0.1	33.1 ± 0.4	0.4 ± 0.1	0.54	1.36	18.1 ± 0.2

A_d, ash content (dry basis); V_d, volatile content (dry basis); FC_d, fixed carbon content (dry basis).
RH, raw rice husk; RH150, wet torrefied rice husk at the torrefaction temperature of 150 °C; RH180, wet torrefied rice husk at the torrefaction temperature of 180 °C; RH210, wet torrefied rice husk at the torrefaction temperature of 210 °C; RH240, wet torrefied rice husk at the torrefaction temperature of 240 °C.

Table 2
Concentration of typical alkali and alkaline earth metal species (AAEMs) in rice husk samples.

Samples	AAEMs concentration (mg/kg, db)			
	K	Na	Ca	Mg
RH	3980 ± 150	256 ± 30	1584 ± 85	621 ± 35
RH150	450 ± 30	54 ± 4	650 ± 32	120 ± 11
RH180	413 ± 20	52 ± 8	550 ± 23	122 ± 8
RH210	390 ± 15	35 ± 4	352 ± 21	105 ± 7
RH240	292 ± 12	20 ± 3	334 ± 15	95 ± 7

In a word, wet torrefaction not only improved the fuel characteristics but also removed a large amount of AAEMs with the dual advantages of dry torrefaction and demineralisation.

3.1.3. FTIR analysis of functional groups

The FTIR spectra of raw and wet torrefied samples are shown in Fig. S1 to investigate the changes in functional groups during wet torrefaction. It contained several similar absorption peaks, such as the strong and broad peak appearing at 3450 cm^{-1} for O—H stretching vibration of hydroxyl and carboxyl groups; the peak at 2930 cm^{-1} for C—H stretching vibration in aliphatic and hydroaromatic structures; the peak positioned at 1730 cm^{-1} for C=O vibration in relation to carbonyl and ester groups from hemicellulose; the peak located at 1610 cm^{-1} for C=C stretching ascribed to aromatic groups; the peaks centered at 1080 and 460 cm^{-1} for Si—O—Si and Si—O stretching due to the presence of SiO_2 [23,32]. Obviously, the absorption intensity at 3440 and 1730 cm^{-1} dramatically weakened due to dehydration and decarboxylation reactions by wet torrefaction.

3.1.4. Surface morphology analysis

The surface morphologies of rice husk samples used to investigate the wet torrefaction are presented in Fig. S2. It can be seen that RH showed a ship-like structure with a organized corrugated surface. With the increase in severity of wet torrefaction, the shrinkage and cracking can be observed for rice husk sample, which was ascribed to the removal of hemicellulose and part of cellulose during wet torrefaction [13]. Meanwhile, applying wet torrefaction has the effects on the reduction of particle size of the rice husk samples. In addition, for rice husk samples of RH210 and RH240, numerous microparticles were found adhering to the surface of rice husk samples, which were thought to be derived from polymerization of hydrolysis and dissolution products during wet torrefaction [33].

3.2. Effects of wet torrefaction on pyrolysis product properties

3.2.1. Pyrolysis product yields

Fig. 4(a) shows the biochar, bio-oil and non-condensable gas yields based on pretreated rice husk samples resulting from pyrolysis of RH, RH150, RH180, RH210 and RH240. For raw RH, the yields of biochar, bio-oil and non-condensable gas were 29.8%, 38.2% and 32.0%, respectively. When applying wet torrefaction at the reaction temperature of 150 °C, the bio-oil yields were obviously higher than that of raw RH, such as 45.4% for RH150. Meanwhile, the yields of biochar and non-condensable gas were reduced from 29.8% to 27.4% for biochar and 32.0% to 27.2% for non-

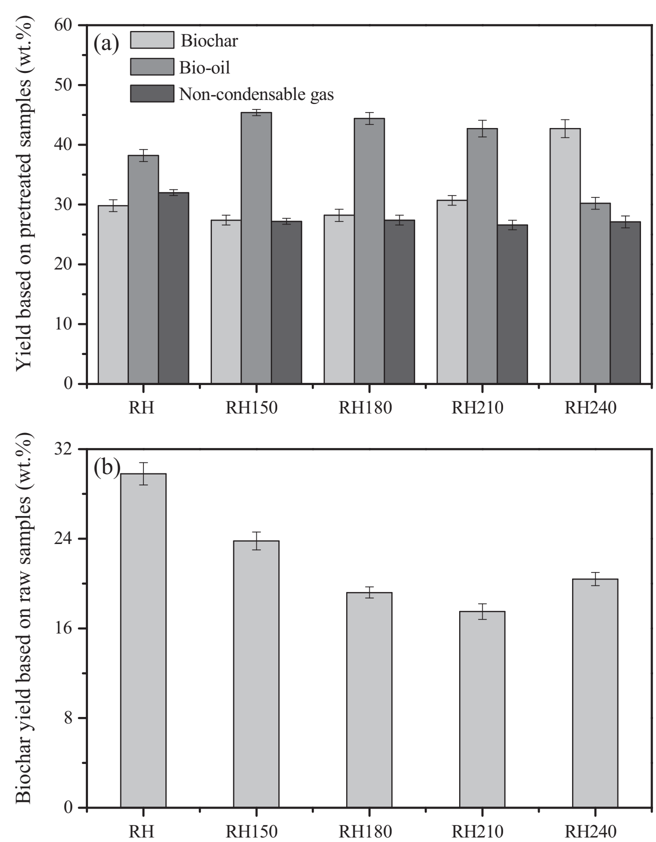

Fig. 4. (a) Pyrolysis product yields (average of three replications ± standard deviations) based on pretreated rice husk samples; (b) biochar yield (average of three replications ± standard deviations) based on raw rice husk samples.

condensable gas in comparison with that of raw RH. This can be attributed to the reduction on AAEMs content and its catalytic effects on pyrolysis pathways. Further enhancing the severity of wet torrefaction from 150 to 210 °C, the bio-oil yields were slightly reduced from 45.4% for RH150 to 42.7% for RH210, and the biochar yields were increased from 27.4% for RH150 to 30.7% for RH210. However, the bio-oil yield was significantly reduced to 30.2% and the biochar yield was enhanced to 42.7% when the wet torrefaction temperature was increased to 240 °C. This can be explained that as the result of the increase in the relative content of lignin, which produced more biochar than the hemicellulose and cellulose during pyrolysis [34]. As shown in Fig. 4(b), the biochar yields based on raw rice husk samples were obviously lowered by wet torrefaction at the reaction temperature from 150 to 210 °C. This result may be explained by less carbonization and cross-linking reaction at relatively mild reaction conditions during wet torrefaction [14]. In general, wet torrefaction at the mild reaction condition has a positive impact on bio-oil production, but more severe wet torrefaction condition was not reasonable.

3.2.2. Bio-oil analysis

The main chemical species in bio-oil were determined by GC–MS, and the relative content of each component calculated by the chromatographic area percentage using the semi-quantitative method was listed in Table 3. According to the results, they can be classified into acids, ketones, furans, phenols, and sugars. It can be observed that there was a large change on the relative content of the chemical species in bio-oil after wet torrefaction. The relative contents of acids, ketones, furans, phenols and sugars for raw RH were 7.4%, 4.3%, 11.0%, 49.2% and 7.7%, respectively. After wet torrefaction, the relative contents of acids, ketones, furans and phenols decreased, while there was an obvious increase in the sugars (mainly levoglucosan) content. The highest relative content of levoglucosan (57.2%) was obtained for RH210, which was about 6.2 times higher than that of raw RH. RH240 has a relatively low content of levoglucosan (40.6%) compared to that of RH210, which may be attributed to the removal of some cellulose from rice husk samples at more severe wet torrefaction condition.

Wet torrefaction removed selectively hemicellulose and ash (AAEMs) from rice husk samples, which can explain the changes on the chemical species in bio-oil. The removal of hemicellulose by wet torrefaction not only effectively reduced the formation of most acids, ketones and furans that derived from hemicellulose pyrolysis in bio-oil, but also weakened the interactions between hemicellulose and cellulose. It has been reported that this interactions promoted the formation of furans at the expense of sugars [35]. In addition, wet torrefaction enhanced the crystallinity degree of cellulose, resulting in the high levoglucosan selectivity during cellulose pyrolysis [36]. Previous reports have indicated that the presence of AAEMs especially K hampered the production of levoglucosan from cellulose pyrolysis and enhanced rearrangement, dehydration and fragmentation reactions [37].

In addition to the changes in the distribution of chemical species of bio-oil by wet torrefaction, the physical properties of bio-oil were also investigated. As shown in Fig. 5, the water content of bio-oil decreased gradually with the increase in wet torrefaction temperature, and it was obviously lower than that from raw RH. The removal of AAEMs and hemicellulose from rice husk samples reduced the formation of water by dehydration reactions during pyrolysis [20]. Wet torrefaction also gave rise to an increase in HHVs of bio-oil in comparison with that of RH. Due to the reduction in the acids production in bio-oil from wet torrefied sample, the pH values were found to show a slight rising trend from 2.3 for RH to 2.7 for RH240. According to the results of the chemical species and physical properties of bio-oil, it was found that wet torrefaction significantly enhanced the bio-oil quality.

Table 3
Relative content of main chemical species in bio-oil obtained from raw and wet torrefied rice husk samples.

Compound	Formula	Relative content (area %)				
		RH	RH150	RH180	RH210	RH240
Acids		7.4	1.7	1.0	0.5	0.7
Acetic acid	$C_2H_4O_2$	6.7	1.6	0.7	0.4	0.3
n-Hexadecanoic acid	$C_{16}H_{32}O_2$	0.7	0.1	0.3	0.1	0.4
Ketones		4.3	3.2	1.0	1.8	1.6
1-Hydroxy-2-butanone	$C_4H_8O_2$	0.8	0.5	0.2	0.3	0.2
Cyclopentanone	C_5H_8O	0.3	0.5	0.2	0.4	0.2
1,2-Cyclopentanedione	$C_5H_6O_2$	0.8	1.0	0.4	0.6	0.6
1,2-Cyclopentanedione, 3-methyl-	$C_6H_8O_2$	2.4	1.2	0.2	0.5	0.6
Furans		11.0	7.2	2.6	2.8	3.8
Furfural	$C_5H_4O_2$	1.8	1.1	0.9	1.0	0.8
2-Furanmethanol	$C_5H_6O_2$	1.1	0.6	0.3	0.4	0.3
Furan, 2,5-diethoxytetrahydro-	$C_8H_{16}O_3$	8.1	5.5	1.4	1.4	2.7
Phenols		49.2	38.2	29.4	23.1	30.1
Phenol, 2-methoxy-	$C_7H_8O_2$	5.2	3.4	1.0	1.3	1.9
Phenol, 4-methyl-	C_7H_8O	2.2	1.6	0.4	0.7	0.8
Phenol, 2,3-dimethyl-	$C_8H_{10}O$	0.8	0.4	0.2	0.4	0.3
Phenol, 2-methoxy-4-methyl-	$C_8H_{10}O_2$	5.1	5.2	3.6	5.5	7.8
Benzofuran, 2,3-dihydro-	C_8H_8O	7.5	6.3	4.3	1.9	2.0
Phenol, 4-ethyl-2-methoxy-	$C_9H_{12}O_2$	6.4	3.8	4.29	3.2	4.9
2-Methoxy-4-vinylphenol	$C_9H_{10}O_2$	8.2	5.9	5.0	2.9	3.8
Phenol, 2-methoxy-	$C_7H_8O_2$	5.2	3.4	1.0	1.3	1.9
Phenol, 4-methyl-	C7H8O	2.2	1.6	0.4	0.7	0.8
Eugenol	$C_{10}H_{12}O_2$	1.6	1.3	1.7	1.3	1.7
Vanillin	$C_8H_8O_3$	0.9	1.1	2.4	0.9	1.0
Phenol, 2-methoxy-4-(1-propenyl)-, (E)-	$C_{10}H_{12}O_2$	2.4	3.0	3.6	1.8	2.0
Phenol, 2,6-dimethoxy-4-(2-propenyl)-	$C_{11}H_{14}O_3$	1.5	1.2	1.5	1.2	1.2
Sugars		7.7	27.5	38.2	57.2	40.6
Levoglucosan	$C_6H_{10}O_5$	7.7	27.5	38.2	57.2	40.6

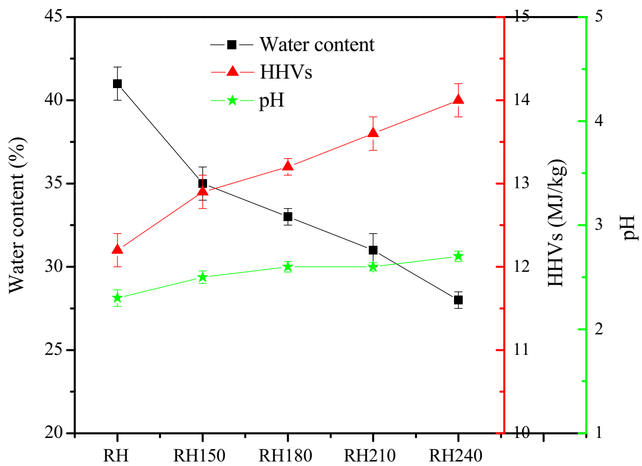

Fig. 5. Physical properties of bio-oil from raw and wet torrefied rice husk samples.

3.2.3. Biochar analysis

The changes on fuel characteristics of biochar by wet torrefaction prior to pyrolysis were also evaluated. The results are shown in Table 4 and revealed that the ash content of biochar from rice husk sample was much higher than that of other biomass char [38]. And the biochar from rice husk sample also has the low values of HHV which were in the range of 18.3–18.8 MJ/kg. High ash content and low HHVs of biochar limited its potential to be used as the solid fuel for heat and electricity production.

It is interesting that the ash content in biochar increased obviously after wet torrefaction. To further investigate the ash composition in biochar, the ash was prepared by combustion of biochar at the temperature of 815 °C for 2 h [39]. The chemical components of ash in biochar determined by XRF are presented in Table S1. It can be seen that SiO_2 was the main component in ash with the concentrations of 90.3–98.7%. Other inorganic oxides, such as K_2O, Na_2O, CaO, MgO, SO_3, P_2O_5, Fe_2O_3, Al_2O_3 and the trace amounts of heavy metals can be also found in the ash. The concentration of SiO_2 increased from 90.3% for RHC to 98.7% for RHC240 with the increase in wet torrefaction severity. It has been proved that Si is very stable in a $SiO_2{\cdot}nH_2O$ form in rice husk and as the result it was hard to be removed by wet torrefaction [31]. This can be used to explain the increase in ash content in biochar from rice husk samples after wet torrefaction and pyrolysis. There was a dramatically reduction in the concentration of K_2O, Na_2O, CaO and MgO in rice husk ash after wet torrefaction, which was consistent with the results of ICP-OES analysis of rice husk samples.

In general, the biochar produced from rice husk samples after wet torrefaction and pyrolysis seemed not a desirable solid fuel. However, taking into account the high ash content in biochar and high SiO_2 concentration in the ash in biochar obtained from wet torrefaction and pyrolysis, this biochar can be used as the feedstock for preparation of nanosilica [40,41].

4. Conclusions

The effects of wet torrefaction on the physicochemical properties and pyrolysis product properties of rice husk sample were investigated in this work. Wet torrefaction has significant changes on the physicochemical properties of rice husk sample. It can be seen that wet torrefaction not only improved the fuel characteristics but also removed a large amount of AAEMs with the dual advantages of dry torrefaction and demineralisation. The results of pyrolysis experiment revealed that wet torrefaction at the mild reaction condition had a positive impact on bio-oil yield, but more severe wet torrefaction condition was not reasonable. The bio-oil quality was improved with the low water content and high heating values when wet torrefaction has been applied prior to pyrolysis process. Meanwhile, the high yield of levoglucosan in bio-oil was obtained from pyrolysis of rice husk sample after wet torrefaction. In addition, the biochar produced from rice husk sample after wet torrefaction and pyrolysis seemed not a desirable solid fuel, it can be used as the feedstock for preparation of nanosilica.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.enconman.2016.10.002.

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Table 4
Fuel characteristics of biochar produced from rice husk samples.

Samples	Proximate analysis (wt.%, db)			Ultimate analysis (wt.%, db)				HHV (MJ/kg)
	A_d	V_d	FC_d	C	H	O	N	
RHC	36.5 ± 0.4	14.5 ± 0.3	49.0 ± 0.4	50.9 ± 0.5	2.0 ± 0.1	10.1 ± 0.2	0.5 ± 0.1	18.4 ± 0.2
RHC150	37.4 ± 0.3	14.6 ± 0.2	48.0 ± 0.4	50.0 ± 0.4	2.1 ± 0.1	10.0 ± 0.3	0.5 ± 0.1	18.8 ± 0.2
RHC180	39.2 ± 0.4	15.1 ± 0.3	45.7 ± 0.4	48.5 ± 0.4	1.8 ± 0.1	10.0 ± 0.2	0.5 ± 0.1	18.3 ± 0.1
RHC210	38.9 ± 0.5	15.7 ± 0.2	45.4 ± 0.5	48.3 ± 0.5	1.9 ± 0.1	10.5 ± 0.3	0.4 ± 0.1	18.3 ± 0.1
RHC240	38.5 ± 0.3	15.9 ± 0.2	45.6 ± 0.4	48.8 ± 0.3	2.1 ± 0.1	10.2 ± 0.2	0.4 ± 0.1	18.4 ± 0.2

A_d, ash content (dry basis); V_d, volatile content (dry basis); FC_d, fixed carbon content (dry basis).
RHC, biochar from raw rice husk; RHC150, biochar from wet torrefied rice husk at the torrefaction temperature of 150 °C; RHC180, biochar from wet torrefied rice husk at the torrefaction temperature of 180 °C; RHC210, biochar from wet torrefied rice husk at the torrefaction temperature of 210 °C; RHC240, biochar from wet torrefied rice husk at the torrefaction temperature of 240 °C.

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